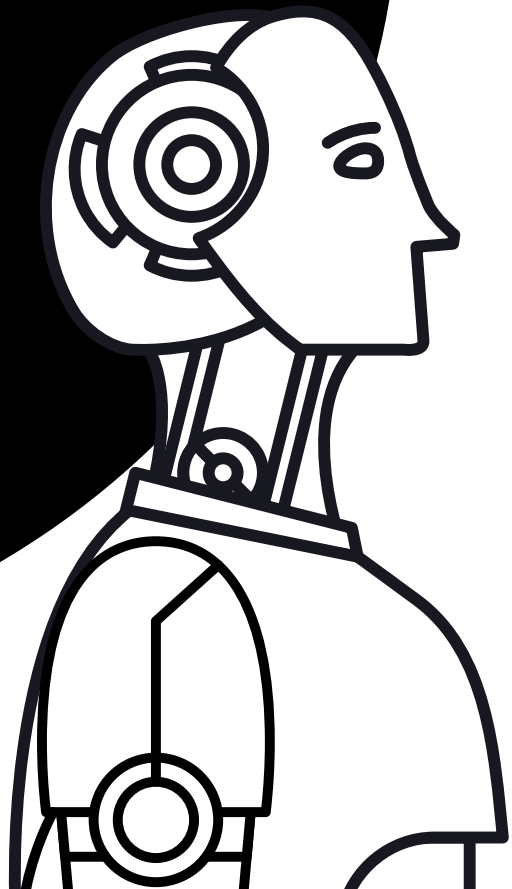


STUDENT
PERFORMANCE
ANALYSIS
DATA SCIENCE

WIKTOR WŁODARCZYK
ROSTYSLAV DMYTROVSKYI



GOAL & DATASET

01

395 STUDENTS, 33 ATTRIBUTES
(PORTUGUESE SECONDARY
SCHOOLS, MATHS)

02

TARGET: FINAL GRADE G3 (0-20);
PASS = $G3 \geq 10$

03

GOAL: FIND RISK FACTORS, TEST
HYPOTHESES, PREDICT AND
SEGMENT STUDENTS



DATA PREPROCESSING & EDA

01

CLEAN DATA: NO MISSING
VALUES OR DUPLICATES

02

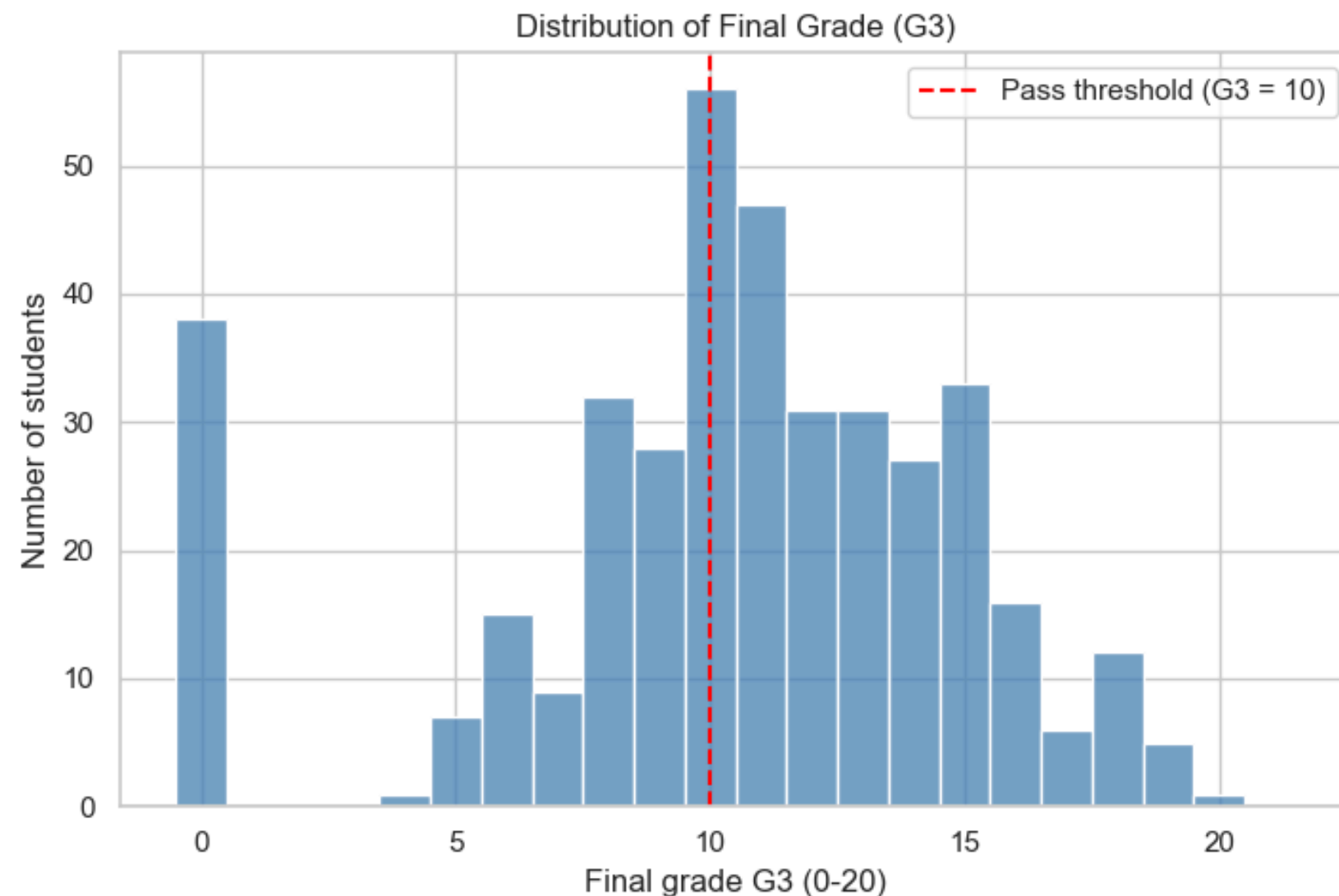
10% OF STUDENTS HAVE $G3 = 0$
(PROBABLE DROPOUTS)

03

EARLIER GRADES DOMINATE: $G2$
VS $G3$ $R = 0.91$

04

FAILURES IS THE STRONGEST
NON-GRADE FACTOR ($R = -0.36$)



HYPOTHESIS TESTING



01

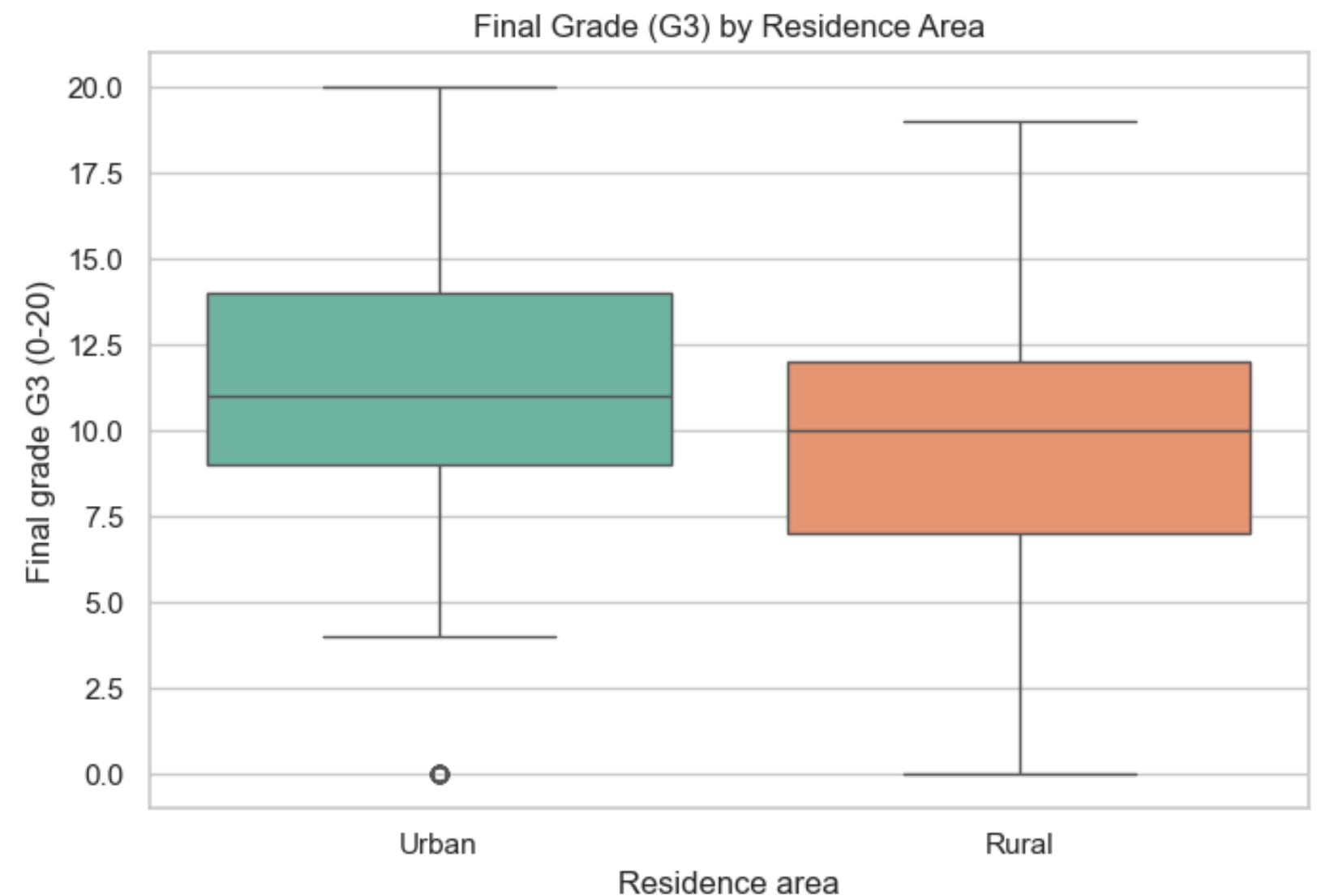
URBAN STUDENTS SCORE
HIGHER: 10.7 VS 9.5 (P = 0.037)

02

EFFECT IS SMALL (COHEN'S D = 0.25)
CONFIRMED BY MANN-WHITNEY

03

NOT SIGNIFICANT AFTER
BONFERRONI CORRECTION
EXPLORATORY FINDING



PREDICTING THE FINAL GRADE

01

RANDOM FOREST IS THE BEST
MODEL: TEST RMSE 1.56, $R^2 = 0.88$

02

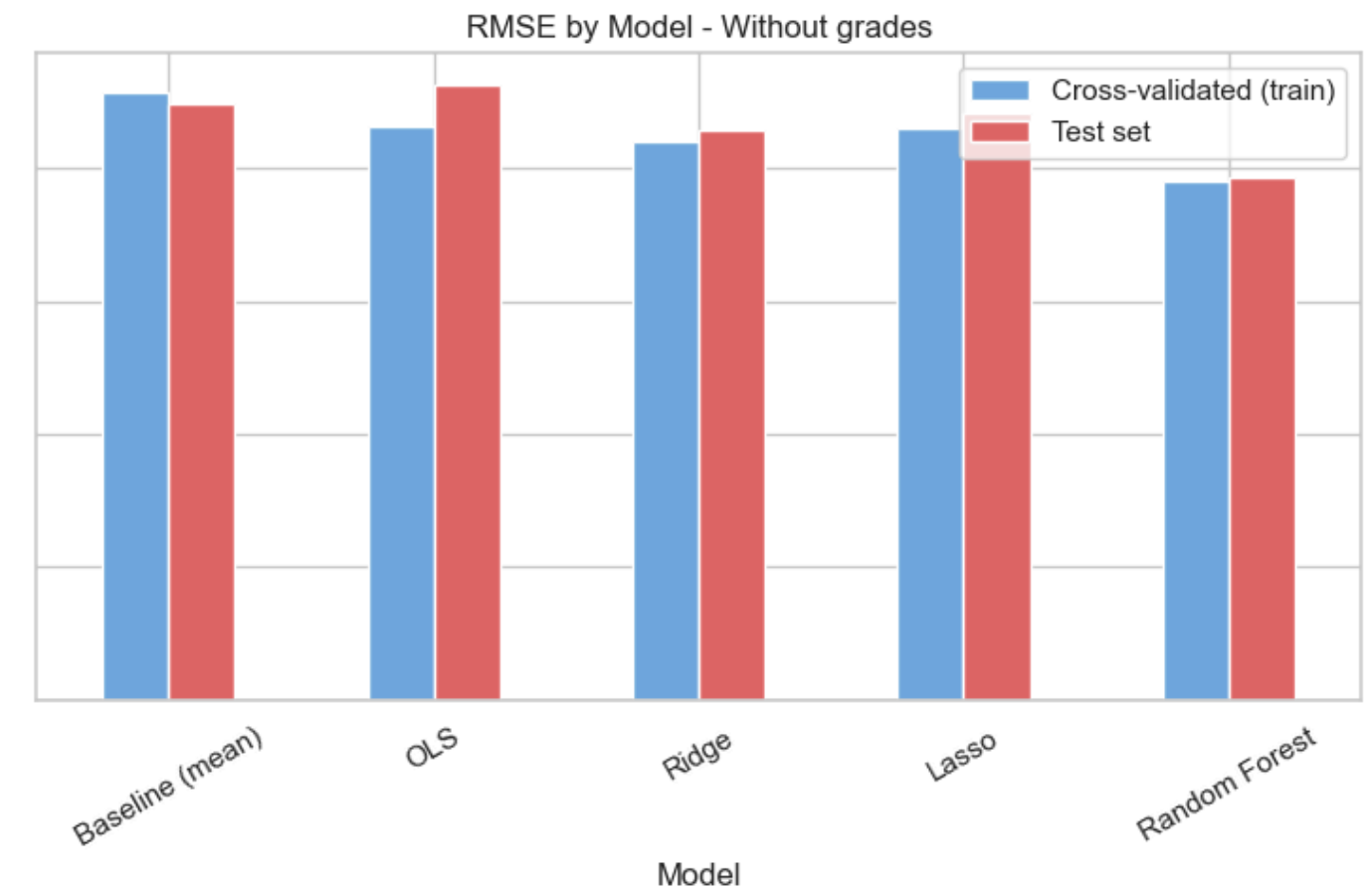
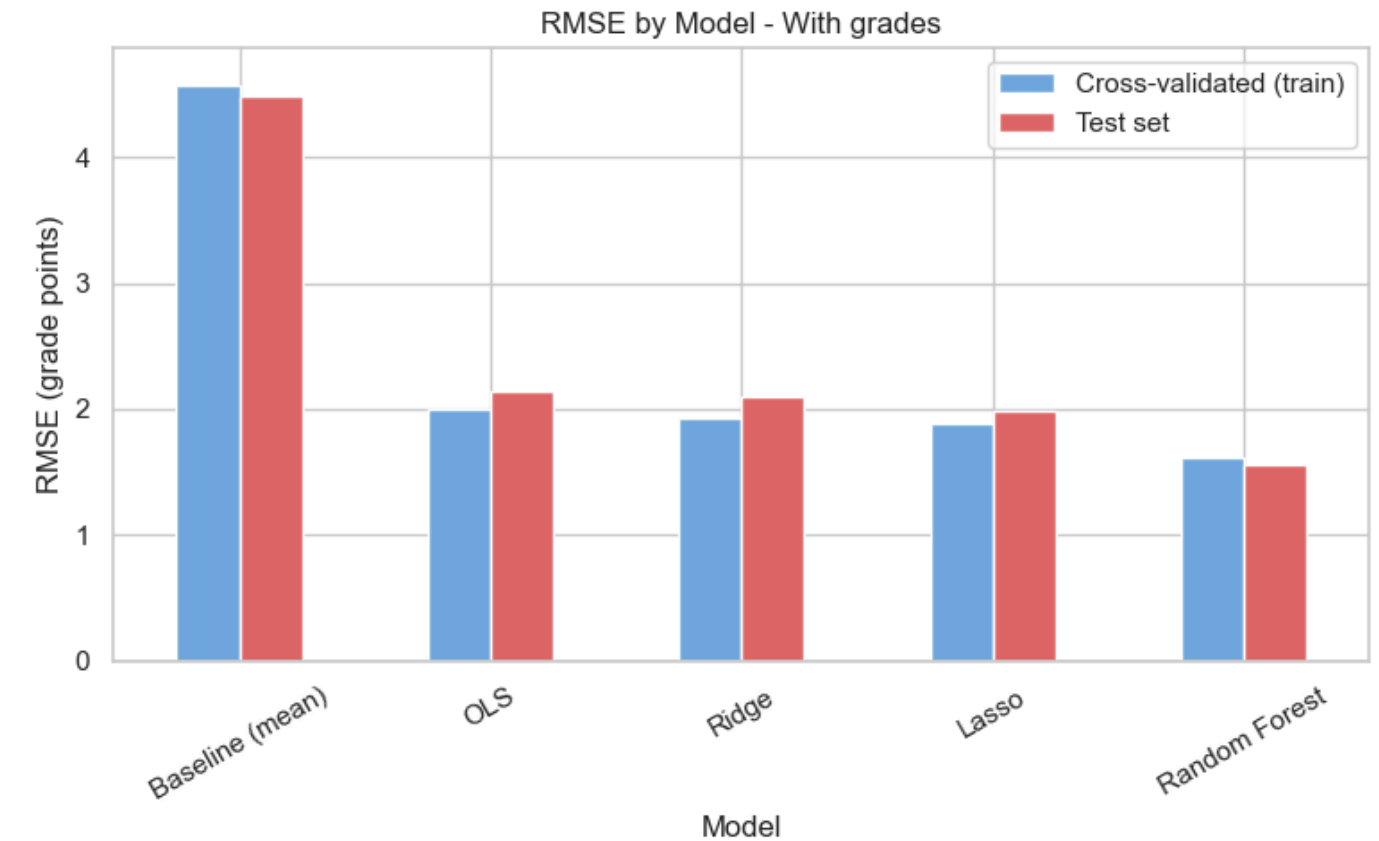
LINEAR MODELS ARE WEAKER (OLS
RMSE 2.14); THE BASELINE IS 4.48

03

WITHOUT G1/G2, R^2 DROPS TO
ABOUT 0.23

04

BACKGROUND FEATURES ALONE
ARE WEAK PREDICTORS



CLASSIFICATION & CLUSTERING

01

PASS/FAIL TREE WITHOUT GRADES: ONE SPLIT ON FAILURES, ACCURACY 0.67 = MAJORITY BASELINE

02

WITH G1/G2: ACCURACY 0.87, FAIL RECALL 0.96

03

K-MEANS (K = 3) FINDS THREE STUDENT TYPES

04

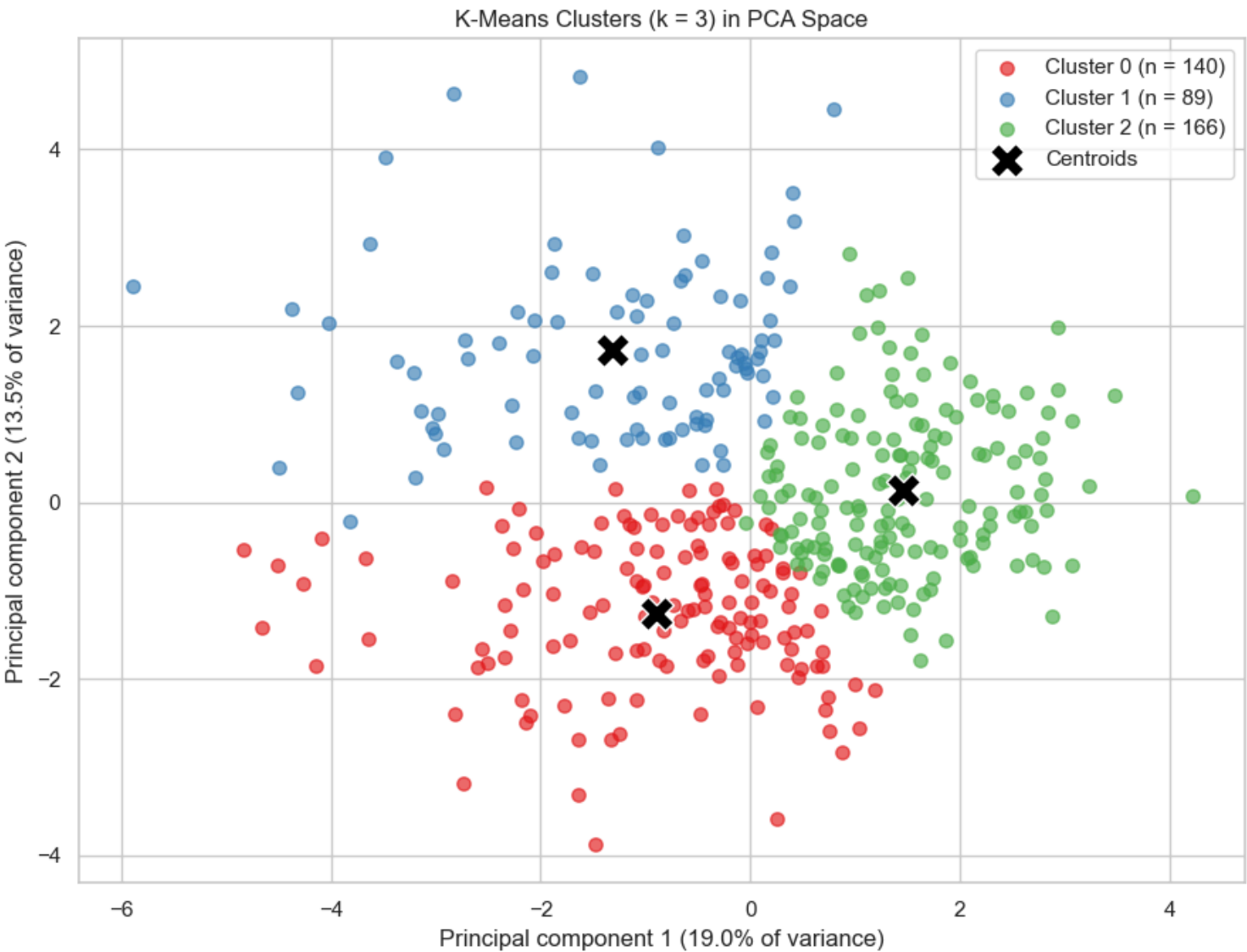
STRUGGLING (35%): MEAN G3 7.4, PASS RATE 44%

05

SOCIAL (23%): 9.9, 61%

06

ACHIEVERS (42%): 13.3, 90%



	Accuracy	Balanced accuracy	Fail recall	Fail precision	F1 (Pass)	ROC-AUC
Without G1/G2 (depth=1, leaf=30)	0.671	0.588	0.346	0.500	0.772	0.588
With G1/G2 (depth=1, leaf=1)	0.873	0.896	0.962	0.735	0.898	0.896

KEY TAKEAWAYS

01

EARLY GRADES
(G1, G2) ARE THE
BEST WARNING
SIGNAL

02

PAST FAILURES
FLAG AT-RISK
STUDENTS EVEN
BEFORE GRADES
EXIST

03

TARGET SUPPORT
AT THE
STRUGGLING GROUP
(MOST FAILURES
AND DROPOUTS)

04

CAVEAT: WEAK
CLUSTER
STRUCTURE,
ABSENCES PARTLY
REFLECTS DROPOUTS

**THANK YOU FOR
YOUR ATTENTION
QUESTIONS?**

GITHUB: [HTTPS://GITHUB.COM/WOXIMUS/DATA-SCIENCE.GIT](https://github.com/woximus/data-science.git)